

因果推断框架下的多因子选股研究

摘要

部分传统 IC 显著因子（如 ROA, ROE）在因果推断中并不显著，存在伪因子风险。因果显著因子包括：MOM60, TURN20。因果显著因子组合样本外夏普（0.0611）高于 IC 显著因子组合（-0.9493），支持因果因子更稳健的判断。

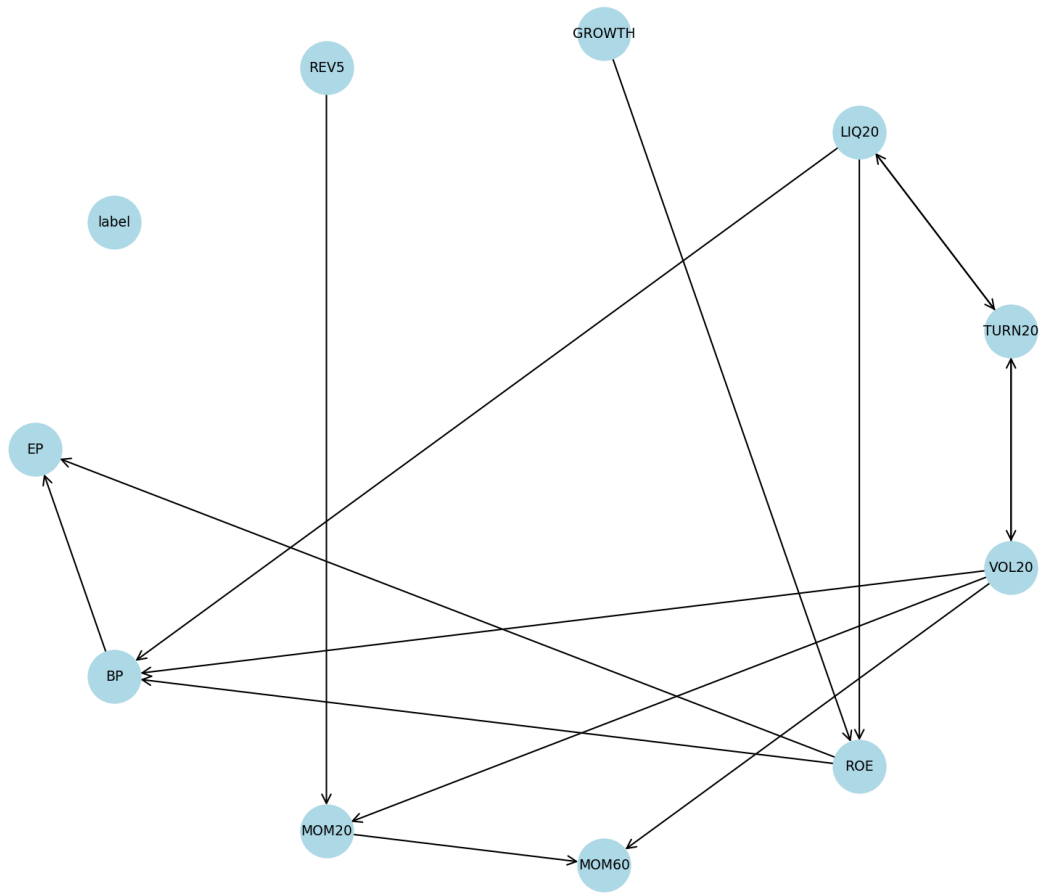
因子定义

因子	含义
EP	盈利收益率 Earnings Yield = 1/PE
BP	账面市值比 Book-to-Price = 1/PB
MOM20	20日动量 (剔除最近1日)
MOM60	60日动量 (剔除最近1日)
ROE	净资产收益率
ROA	总资产收益率
VOL20	20日收益波动率 (年化)
TURN20	20日平均换手率
LIQ20	20日平均成交量 (对数)
GROWTH	净利润同比增长率
REV5	5日短期反转

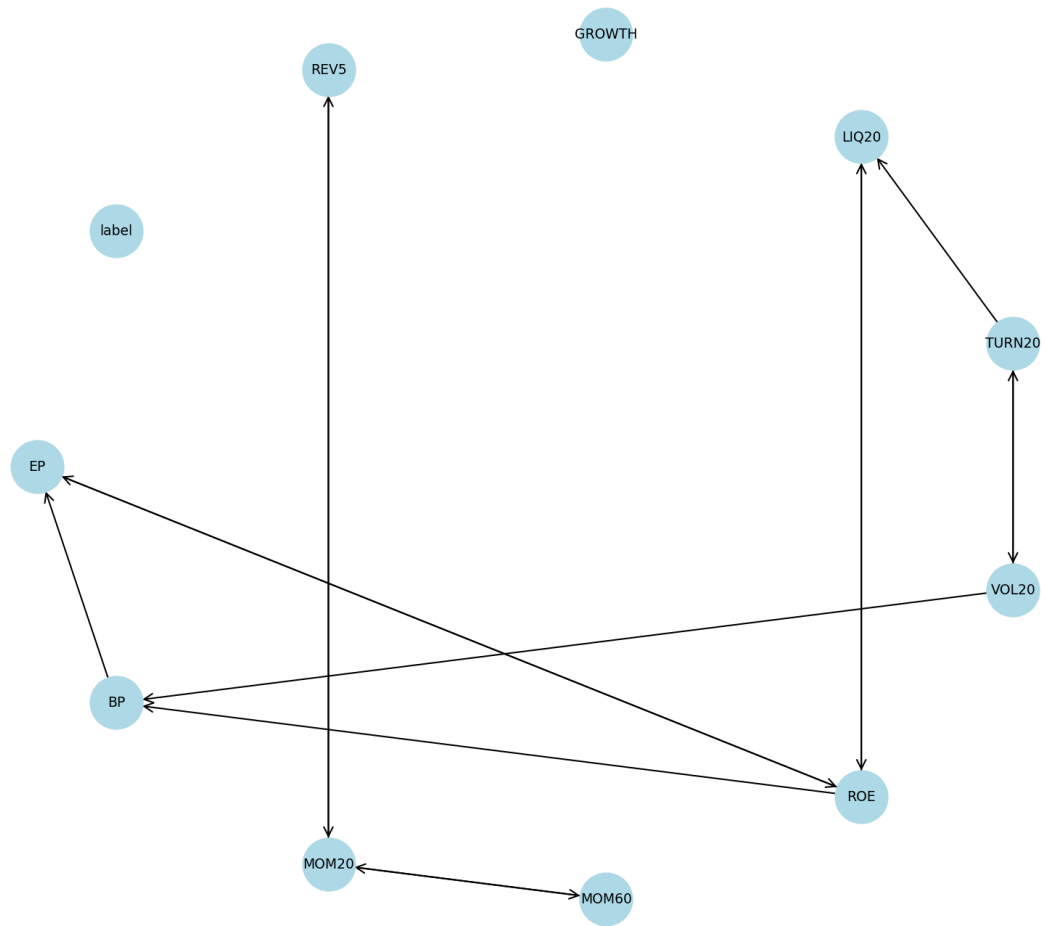
因果发现

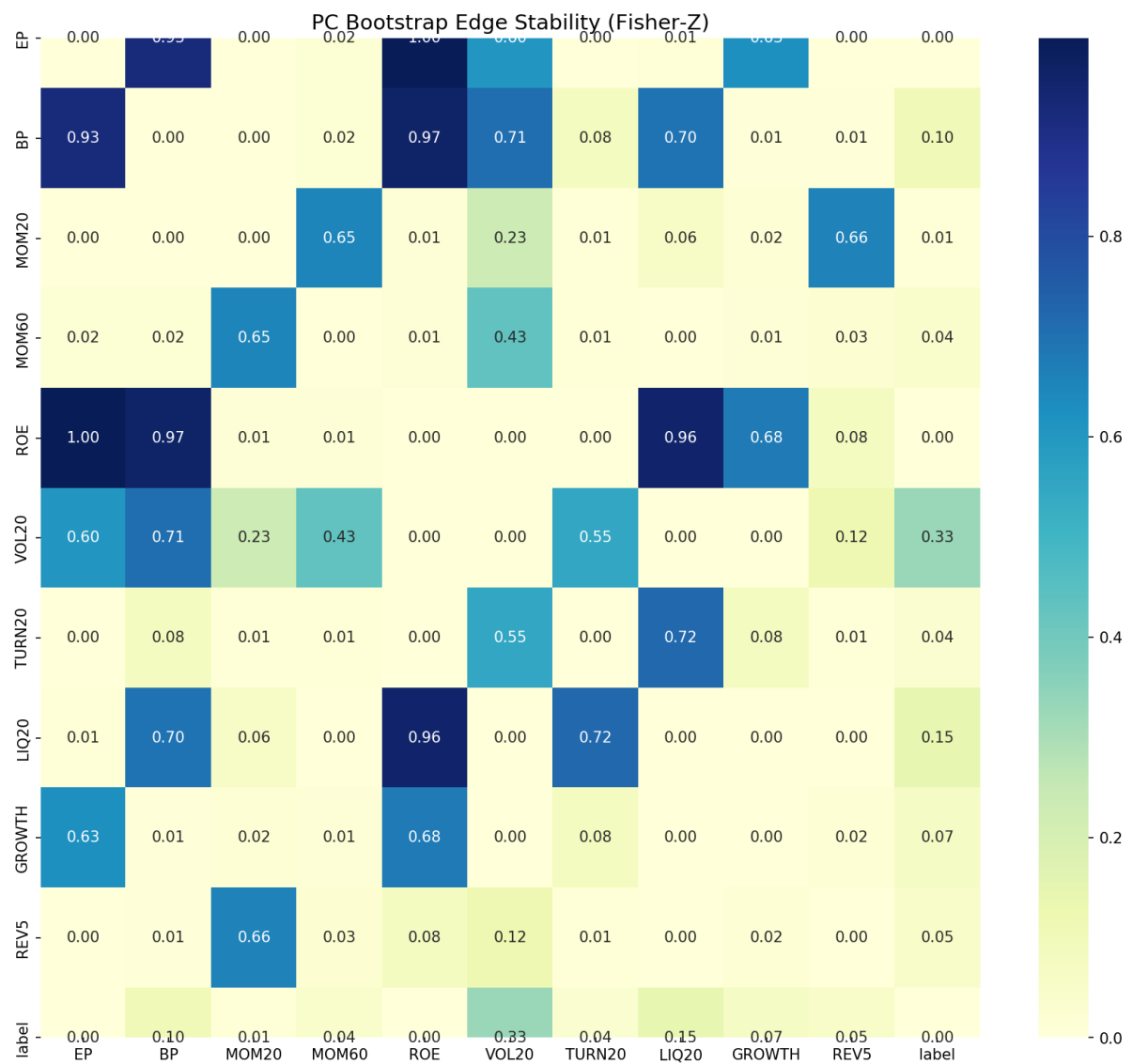
对真实数据进行 **1000 次** Bootstrap 重采样，同时运行 PC 与 FCI 算法。共识别 **6 条** 稳定边（稳定性 > 70%）。

PC (Fisher-Z) CPDAG



FCI (Fisher-Z) PAG





因果效应估计：四维方法矩阵

DML

factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	residDmean	residDstd
EP	DML	-0.00857285	0.00742469	-1.15464	0.248467	-0.0231396	0.00599395	-0.00893832	0.00551476	0.105326	1204	-0.00713237	0.00713237
BP	DML	-0.00222062	0.00665979	-0.333437	0.738863	-0.0152867	0.0108455	0.00591244	0.005904	0.316823	1204	-0.00555109	0.00555109
MOM20	DML	-0.00478444	0.00365921	-1.30751	0.19129	-0.0119636	0.0023947	-0.00182053	0.00378055	0.630214	1204	0.00122843	0.00122843
MOM60	DML	-0.00451168	0.00369654	-1.22051	0.22251	-0.0117641	0.00274071	-0.00619165	0.00377173	0.100939	1204	-0.00873362	0.00873362
ROE	DML	-0.00872118	0.00628298	-1.38807	0.165374	-0.021048	0.00360564	-0.00117537	0.00492518	0.811422	1204	0.00199207	0.00199207
ROA	DML	-0.00872118	0.00628298	-1.38807	0.165374	-0.021048	0.00360564	-0.00117537	0.00492518	0.811422	1204	0.00199207	0.00199207
VOL20	DML	0.00724113	0.00466498	1.55223	0.12087	-0.00191127	0.0163935	0.00773829	0.00427492	0.0705237	1204	-0.00408788	0.00408788
TURN20	DML	0.0201815	0.00452898	4.45609	9.12526e-06	0.0112959	0.0290671	0.0170252	0.00464027	0.000254276	1204	-0.0068857	0.0068857
LIQ20	DML	-0.00323221	0.0044942	-0.719195	0.47216	-0.0120496	0.00558515	-0.00395385	0.0043787	0.366724	1204	-0.00157423	0.00157423
GROWTH	DML	0.00104847	0.00341233	0.30726	0.758698	-0.00564631	0.00774326	0.00276404	0.0034861	0.428009	1204	-0.0169922	0.0169922
REV5	DML	-0.00014071	0.0032584	-0.0431838	0.965562	-0.0065335	0.00625208	0.00045455	0.00341556	0.894151	1204	-0.0122344	0.0122344

IV

factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	residDmean	residDstd
EP	IV	-0.00990085	0.00636159	-1.55635	0.119978	-0.0223862	0.00258448	nan	nan	nan	914	nan	nan

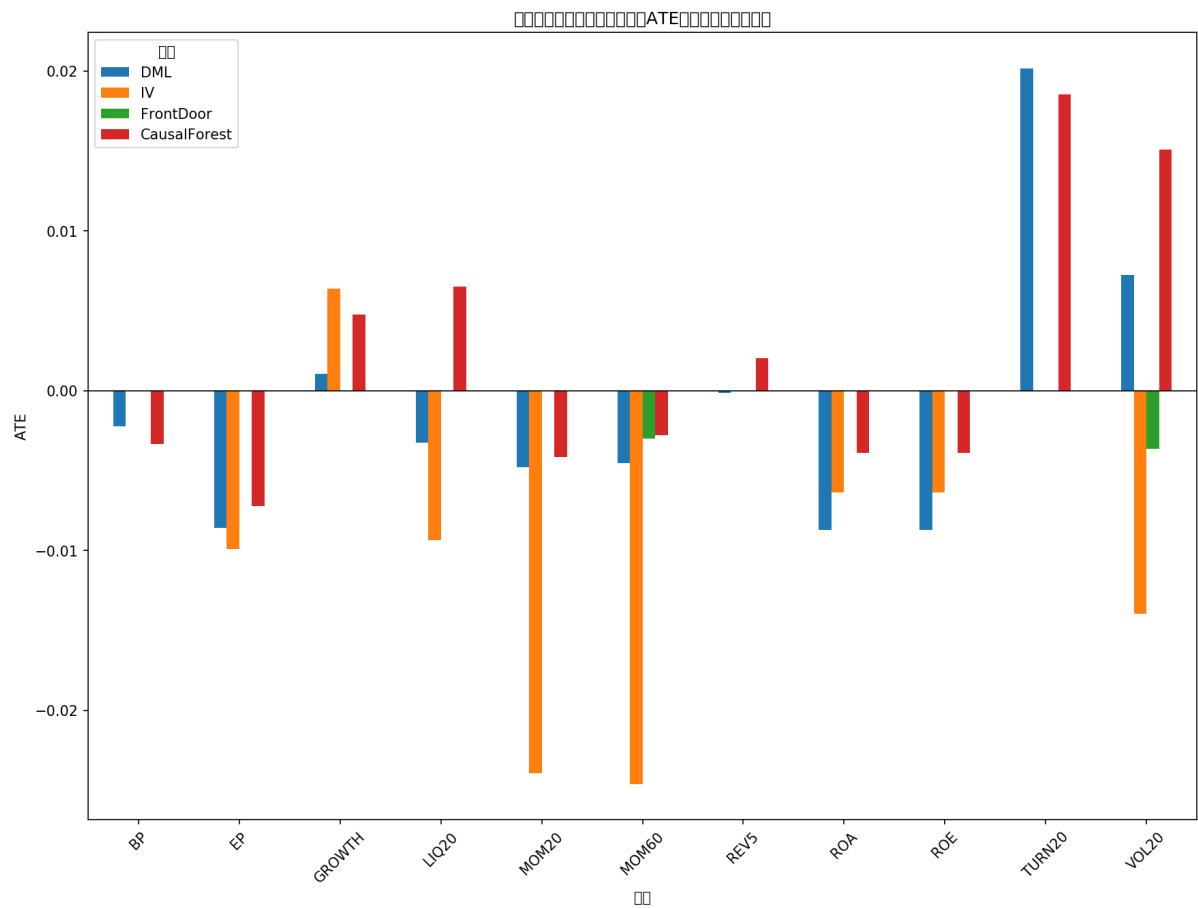
factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	residDmean	residDstd	ci_lower	ci_upper
BP	IV	0.00689673	0.00655677	1.05185	0.293152	-0.00597166	0.0197651	nan	nan	nan	914	nan	nan		
MOM20	IV	-0.0239336	0.00907722	-2.63667	0.0085171	-0.0417487	-0.00611859	nan	nan	nan	914	nan	nan		
MOM60	IV	-0.0246104	0.00582484	-4.22508	2.63266e-05	-0.0360423	-0.0131785	nan	nan	nan	914	nan	nan		
ROE	IV	-0.00635726	0.00572819	-1.10982	0.267374	-0.0175995	0.00488496	nan	nan	nan	914	nan	nan		
ROA	IV	-0.00635726	0.00572819	-1.10982	0.267374	-0.0175995	0.00488496	nan	nan	nan	914	nan	nan		
VOL20	IV	-0.0139388	0.0137797	-1.01155	0.312027	-0.040983	0.0131054	nan	nan	nan	914	nan	nan		
TURN20	IV	0.0193613	0.00662027	2.92455	0.00353643	0.00636828	0.0323543	nan	nan	nan	914	nan	nan		
LIQ20	IV	-0.00933803	0.00529876	-1.7623	0.0783584	-0.0197374	0.00106138	nan	nan	nan	914	nan	nan		
GROWTH	IV	0.00639981	0.00473432	1.35179	0.176783	-0.00289183	0.0156914	nan	nan	nan	914	nan	nan		
REV5	IV	0.019815	0.0108833	1.82068	0.0689886	-0.00154469	0.0411747	nan	nan	nan	914	nan	nan		

前门调整

factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	residDmean	residDstd	ci_lower	ci_upper
EP	FrontDoor	-0.001501	0.00219062	-0.685196	0.493353	-0.00579461	0.00279261	nan	nan	nan	1204	nan	nan		
BP	FrontDoor	-0.00145412	0.00234478	-0.62015	0.535277	-0.00604989	0.00314166	nan	nan	nan	1204	nan	nan		
MOM20	FrontDoor	0.00113945	0.0015021	0.758571	0.448258	-0.00180467	0.00408357	nan	nan	nan	1204	nan	nan		
MOM60	FrontDoor	-0.0030085	0.00151023	-1.99208	0.0465882	-0.00596855	-4.8448e-05	nan	nan	nan	1204	nan	nan		
ROE	FrontDoor	0.00184437	0.00195827	0.941835	0.346466	-0.00199384	0.00568258	nan	nan	nan	1204	nan	nan		
ROA	FrontDoor	0.00184437	0.00195827	0.941835	0.346466	-0.00199384	0.00568258	nan	nan	nan	1204	nan	nan		
VOL20	FrontDoor	-0.00361625	0.00171362	-2.1103	0.035039	-0.00697494	-0.000257551	nan	nan	nan	1204	nan	nan		
TURN20	FrontDoor	-0.000299802	0.0018414	-0.162812	0.870694	-0.00390895	0.00330934	nan	nan	nan	1204	nan	nan		
LIQ20	FrontDoor	-0.000508531	0.00173784	-0.292623	0.769861	-0.00391469	0.00289763	nan	nan	nan	1204	nan	nan		
GROWTH	FrontDoor	0.000732737	0.00138418	0.529364	0.59665	-0.00198026	0.00344574	nan	nan	nan	1204	nan	nan		
REV5	FrontDoor	-0.00128752	0.00135807	-0.948051	0.343294	-0.00394934	0.0013743	nan	nan	nan	1204	nan	nan		

Causal Forest

factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	residDmean	residDstd	ci_lower	ci_upper
EP	CausalForest	-0.00719979	0.00725584	-0.992275	0.321263	-0.0214212	0.00702165	nan	nan	nan	1204	nan	nan		
BP	CausalForest	-0.00331375	0.0072668	-0.456013	0.648463	-0.0175567	0.0109292	nan	nan	nan	1204	nan	nan		
MOM20	CausalForest	-0.00416183	0.00941248	-0.442161	0.658452	-0.0226103	0.0142866	nan	nan	nan	1204	nan	nan		
MOM60	CausalForest	-0.00279208	0.00799013	-0.349441	0.72682	-0.0184527	0.0128686	nan	nan	nan	1204	nan	nan		
ROE	CausalForest	-0.00388849	0.00721611	-0.538862	0.590081	-0.0180321	0.0102551	nan	nan	nan	1204	nan	nan		
ROA	CausalForest	-0.00388849	0.00721611	-0.538862	0.590081	-0.0180321	0.0102551	nan	nan	nan	1204	nan	nan		
VOL20	CausalForest	0.0150985	0.00741533	2.03612	0.0419571	0.000564447	0.0296326	nan	nan	nan	1204	nan	nan		
TURN20	CausalForest	0.0185426	0.00766396	2.41946	0.0156912	0.00352127	0.033564	nan	nan	nan	1204	nan	nan		
LIQ20	CausalForest	0.00652006	0.00863205	0.755331	0.450198	-0.0103988	0.0234389	nan	nan	nan	1204	nan	nan		
GROWTH	CausalForest	0.00476789	0.00748016	0.637404	0.523983	-0.00989323	0.019429	nan	nan	nan	1204	nan	nan		
REV5	CausalForest	0.00204593	0.00868351	0.23561	0.813775	-0.0149738	0.0190656	nan	nan	nan	1204	nan	nan		

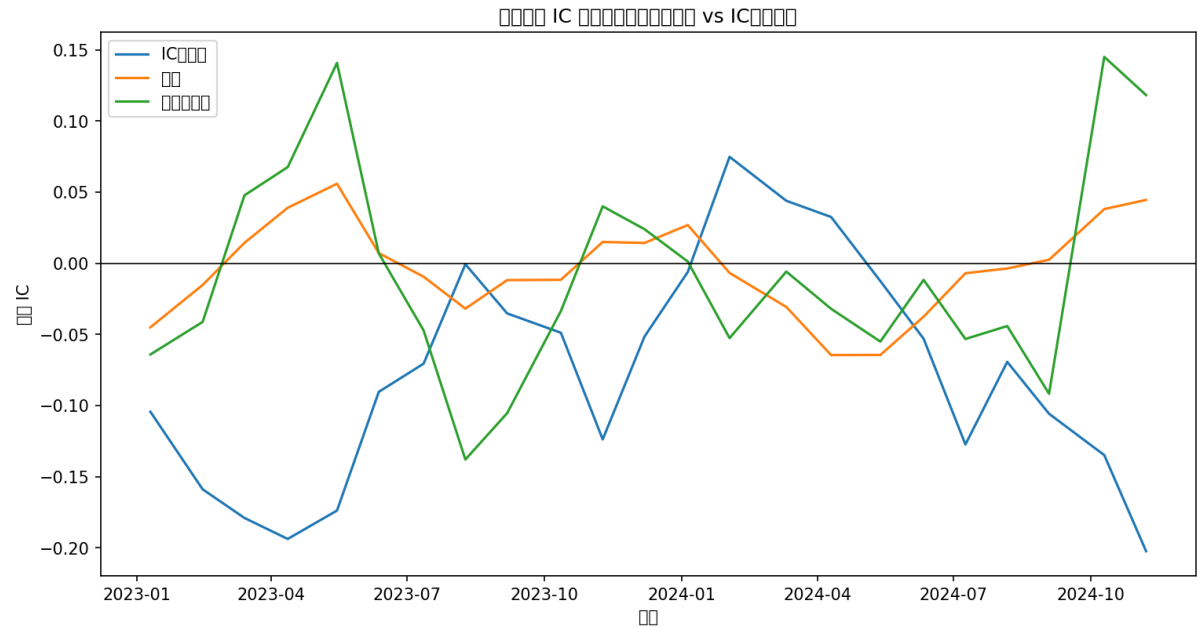


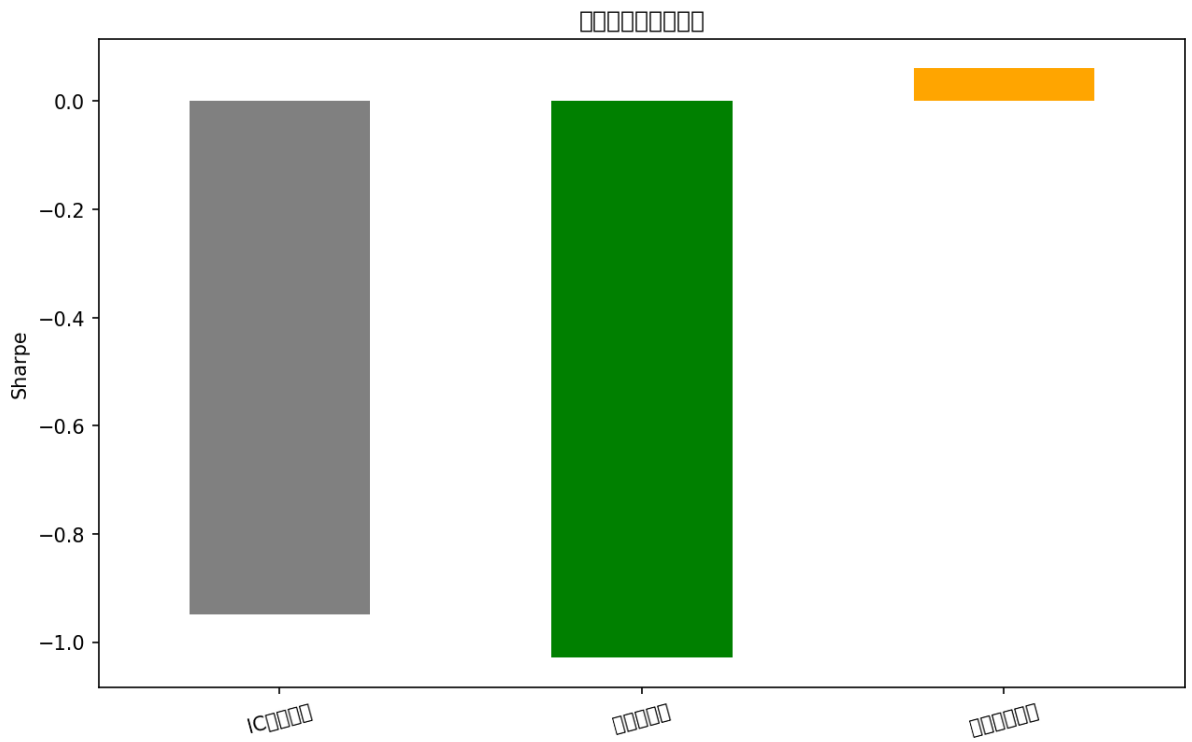
因果因子 vs 相关因子

2×2 分类矩阵

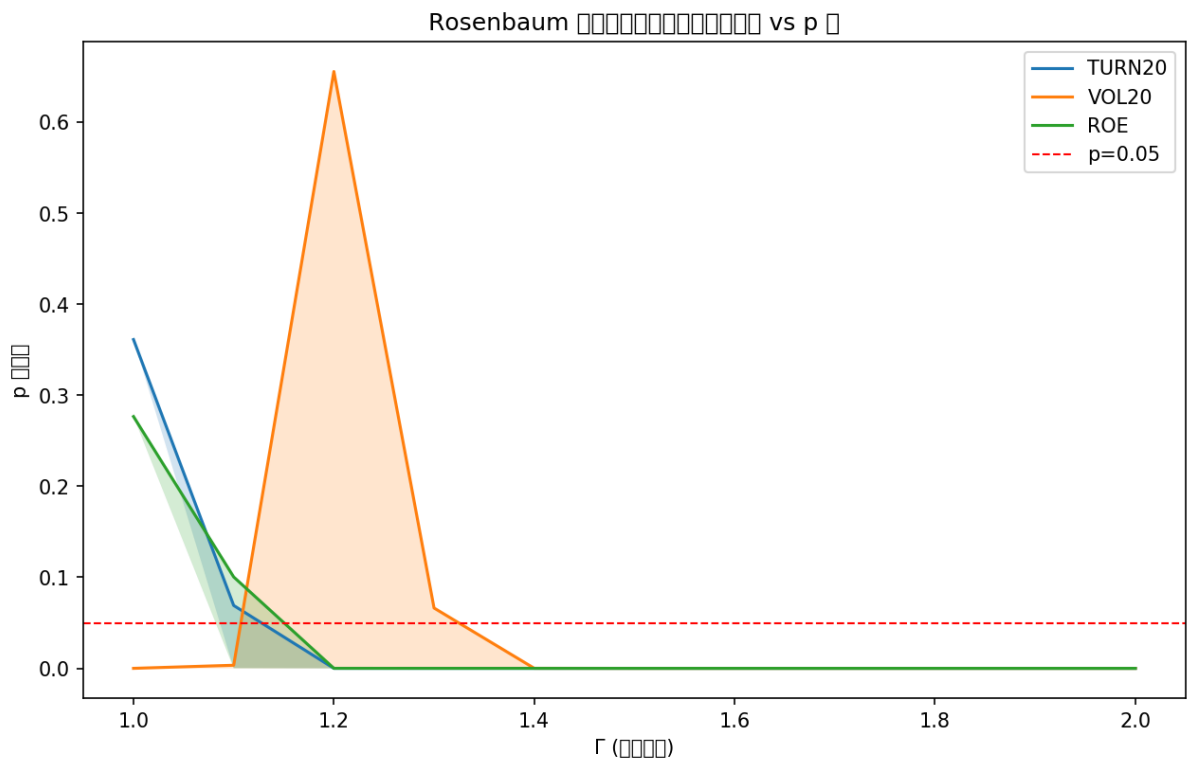
	IC显著	IC不显著
因果显著	0	2
因果不显著	2	7

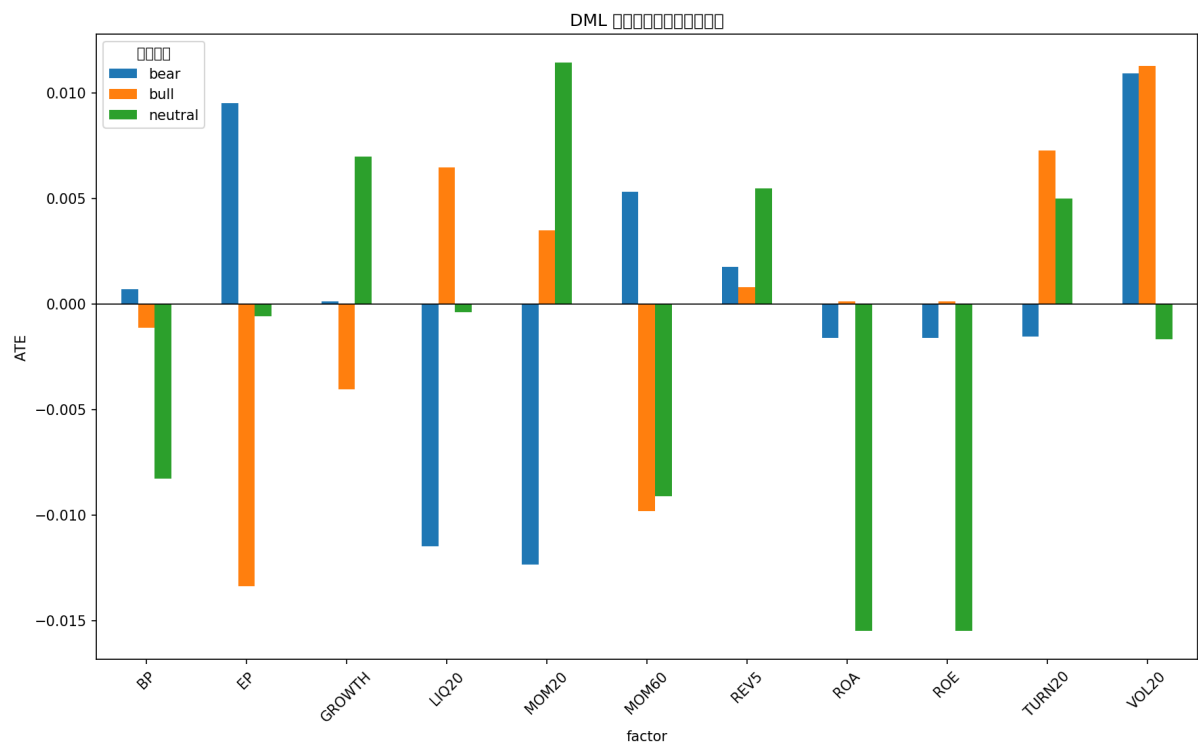
- 因果显著因子: ['MOM60', 'TURN20']
- IC 显著因子: ['ROA', 'ROE']
- 伪因子 (IC 显著但因果不显著): ['ROA', 'ROE']
- 被掩盖的因子 (因果显著但 IC 不显著): ['MOM60', 'TURN20']





稳健性检验





识别假设与数学严谨性

- 平均倾向得分重叠度：0.8849
- 共同支撑评估：良好
- SUTVA 高关注因子：['EP', 'BP', 'MOM20', 'MOM60', 'ROE', 'VOL20', 'TURN20', 'LIQ20', 'REV5']

所有图表保存在 figures/ 目录。正式 PDF 可通过 xelatex report_cn.tex 编译生成。